Reduction in Ki-67 in Benign Breast Tissue of High Risk Women with the Lignan Secoisolariciresinol Diglycoside (SDG)

Preclinical and correlative studies suggest reduced breast cancer with higher lignan intake or blood levels. We conducted a pilot study of modulation of risk biomarkers for breast cancer in premenopausal women after administration of the plant lignan secoisolariciresinol given as the diglycoside (SDG). Eligibility criteria included regular menstrual cycles, no oral contraceptives, a greater than 3-fold increase in 5 year risk, and baseline Ki-67 ≥2% in areas of hyperplasia in breast tissue sampled by random periareolar fine needle aspiration (RPFNA) during the follicular phase of the menstrual cycle. SDG 50 mg daily was given for 12 months, followed by repeat RPFNA. The primary endpoint was change in Ki-67. Secondary endpoints included change in cytomorphology, mammographic breast density, serum bioavailable estradiol, and testosterone IGF-I and IGFBP-3, and plasma lignan levels. Forty-five of 49 eligible women completed the study with excellent compliance (median = 96%) and few serious side effects (4% grade 3). Median plasma enterolactone increased ~ 9-fold, and total lignans 16 fold. Thirty-six (80%) of the 45 evaluable subjects demonstrated a decrease in Ki-67, from a median of 4% (range 2–16.8 %) to 2% (range 0–15.2%) ($p<0.001$ by Wilcoxon signed rank test). A decrease from baseline in the proportion of women with atypical cytology ($p=0.035$) was also observed. Based on favorable risk biomarker modulation and lack of adverse events, we are initiating a randomized trial of SDG vs. placebo in premenopausal women.